

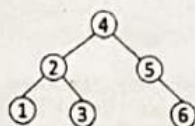
University of Dhaka
Affiliated Engineering Colleges
 B. Sc. in Computer Science & Engineering
 2nd Year 1st Semester Final Examination-2021
CSE 2101 (Data Structures and Algorithms)

Time: 3.00 Hours

[Answer any Five questions]

Marks: 70

1. a) What is Abstract Data Type? Define Data Structure. 6
 b) Briefly describe complexity and space-time tradeoff of algorithms? 4
 c) How to represent linear array in memory. 4
2. a) Why algorithm analysis is important? 2
 b) What is asymptotic analysis of an algorithm? What are asymptotic notations? 4
 c) Analyze the complexity of linear and binary search algorithm. 5
 d) What are the differences between array and linked list? 3
3. a) Simulate the pre-order and in-order tree traversal of the following tree 4



- b) Illustrate similar and copies of a tree with example? 3
 c) Write an algorithm to traverse a linked list? 4
 d) Define overflow, underflow and garbage collection? 3
4. a) Let n denote a positive integer. Suppose a function L is defined recursively as follows: 4

$$L(n) = \begin{cases} 0 & \text{if } n = 1 \\ L[n/2] + 1 & \text{if } n > 1 \end{cases}$$

 What does the function do?
 b) Illustrate stack POP and PUSH operation. 6
 c) Write down the two mandatory properties of a recursive function? Define Queues and Deques. 4
5. a) What is Traversing? What are the different types of traversing? 2+2
 b) Construct a Binary tree for the expression $[(a + b * c)] - [(d * e + f)/(g-h)]$. What will be the outputs for Pre order and Post order of this expression? 6
 c) Write an Algorithm to Insert an element in a sorted Heap. 4
6. a) Write down the complexity of Quick Sort, Bubble Sort, Linear Search and Binary Search Algorithm. 4
 b) Suppose you have an Array of following elements: 44, 22, 77, 33, 99, 11, 66, 55
 Explain the technique for Selection Sort from above elements. 6
 c) Write an algorithm for Insertion Sort. 4
7. a) Distinguish between Linear and Non Linear Data Structure. 2
 b) What is the condition for a Complete Binary Tree? Draw a complete Binary Tree for the elements of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20. 1+3
 c) Explain the technique for Merge Sort Algorithm with example. 4
 d) Draw Max Heap and Mean Heap Tree from the following elements: 50, 20, 70, 40, 30, 60, 90, 80. 4